GERMÁN KÜBER

Senior Rust Engineer — Distributed Systems, Async Runtimes & Low-Latency Infrastructure

Las Palmas, Spain · EU Citizen — authorized to work across the EU/EEA, no visa sponsorship required
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PROFESSIONAL SUMMARY

Senior Rust engineer with 4+ years shipping production tokio-based async Rust in low-latency, high-throughput distributed systems, on top of 15+ years building systems across multiple languages. Designed, shipped, and operated tokio async pipelines processing $50M+ in daily transactional flow with execution under 10ms at Borderless Capital, delivering memory safety and zero-cost abstractions without garbage collection overhead. Strong systems fundamentals across async runtime patterns (tokio runtime tuning, Pin, structured concurrency), low-latency observability, Python / Cython FFI bindings, custom protocol implementations, and reverse-engineering of Go (Geth) internals to inform Rust-side designs. Distributed systems experience spans multi-chain blockchain ecosystems (EVM, Solana, Polkadot / Substrate), consensus protocols (hands-on with GRANDPA / BABE, familiar with Raft), and Zero-Knowledge proof toolchains. Comfortable in adjacent ecosystems (**TypeScript / Node.js, Python, .NET**) and across cloud-native infrastructure on AWS and Azure. Active contributor to the broader systems and Web3 communities through 80+ international talks. EU citizen (Spanish nationality) with full authorization to work across the EU/EEA — no visa or sponsorship required — open to self-funded relocation anywhere in the EU.

CORE EXPERTISE

Rust Core & Ecosystem: Rust (stable, async / await, Pin, Generic Associated Types / GATs, async traits, trait objects, lifetimes, no_std familiar), Cargo workspace management, rustfmt, clippy, criterion (benchmarking), cargo-flamegraph (profiling), cargo-fuzz / libfuzzer (fuzzing), miri (UB detection), serde, thiserror, anyhow, tracing, tracing-subscriber

Async Runtimes & Concurrency: tokio (async runtime, runtime tuning, multi-threaded scheduler, channels, tasks, select!, spawn_blocking), futures, async-trait, pin-project, lock-free data structures (crossbeam, atomics, memory ordering), data parallelism (rayon), structured concurrency patterns, custom executors

Systems Programming & Performance: Memory safety without garbage collection, zero-cost abstractions, ownership and lifetimes, low-latency design (sub-10ms execution), FFI with C / C++ / Python, Cython bindings, custom protocol implementations, performance profiling (flamegraphs, perf), memory profiling, lock-free patterns, cache-friendly data layout

Distributed Systems & Consensus: Async-driven high-throughput pipelines, event-driven architectures, fault tolerance, distributed coordination across multi-region systems, replication patterns, eventual consistency, transaction delivery infrastructure, consensus protocols (hands-on with GRANDPA / BABE on Substrate, familiar with Raft and Paxos)

Networking & Protocols: tokio networking (TCP / UDP), HTTP servers (hyper), gRPC (tonic), custom protocol design, mempool analysis, RPC clients, TLS, parallel broadcasting across endpoints, validator / TPU routing, Linux internals (networking stack, sockets, namespaces, epoll)

Observability & Tracing: Prometheus + custom Rust telemetry, tracing crate with structured logging, OpenTelemetry, distributed tracing across services, real-time metrics design at scale (50+ production metrics)

Multi-Language Interop: Rust ↔ Python via Cython / PyO3 bindings, Rust ↔ C / C++ FFI, reverse-engineering Go (Geth / Go-Ethereum) internals, TypeScript / Node.js and .NET / C# integration, polyglot service architectures

Blockchain & Web3: Polkadot SDK, Substrate, FRAME, XCM, parachains, EVM (Ethereum, Arbitrum, Optimism), Solana, MEV infrastructure, Zero-Knowledge proofs (Noir, Barretenberg, Garaga on StarkNet, Cairo)

Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, FastAPI, CI / CD pipelines, Linux internals (networking, sockets, namespaces), Infrastructure as Code (Terraform, ARM templates)

AI / LLM Systems: LLM orchestration (multi-provider), real-time pipelines, MCP (Model Context Protocol) server and client implementation, LangChain, LangGraph

Programming Paradigms & Design: Object-Oriented Programming (OOP), Functional Programming, design patterns, architecture patterns, data structures, algorithm design, SOLID principles, clean code

Engineering Practices & Leadership: Test-Driven Development (TDD), pair programming, code review, mentoring, technical interviewing, ADRs, DevOps practices, CI / CD, Infrastructure as Code (Terraform), on-call and incident response

PROFESSIONAL EXPERIENCE

Independent AI Consultant — Agent Engineering & Enterprise AI Delivery

Self-employed (autónomo, Spain) — Mar 2026 – Present — Remote, client under NDA

- Forward-deployed AI consultant embedded with an enterprise engineering organization: run discovery with business and technical stakeholders, architect the solution, and own delivery from pilot through production and continuous iteration.
- Design and ship production agentic workflows on LangChain, LangGraph and Microsoft Agent Framework, with eval suites and regression tests gating every change before production.
- Built knowledge graphs (Cognee, Graphiti) over internal repositories, including specialized code graphs that map microservice architecture so agents can query real system structure when making decisions.
- Cut inference cost and latency through semantic caching, agent-level semantic memory, and dynamic model routing across providers.

Lead Architect & Co-Founder

KliverAI (Own Venture, closed Feb 2026) — Mar 2025 – Feb 2026 (full-time) — Remote, Spain

- Architected polyglot platform end-to-end (Rust, Node.js / TypeScript, Python, .NET, Kubernetes, AWS) with multi-provider service orchestration; owned ADRs, service boundaries, and technology selection.
- Rewrote the real-time layer from app-embedded SignalR to Azure SignalR Service with Redis-backed state, scaling concurrent users by 500% while holding sub-200ms latency on audio pipelines; implemented performance-critical components in Rust (tokio, async) for the audio and streaming layer.
- Benchmarked TTS / STT services (ElevenLabs, Deepgram, AssemblyAI, OpenAI Whisper, Play.ht) on latency, cost, and quality — getting latency to 90ms and word accuracy to nearly 95%.

- Implemented MCP (Model Context Protocol) servers and clients to standardize service-to-service tool integration patterns.
- Designed a Zero-Knowledge proof system using Noir, Barretenberg, and Garaga on StarkNet for verifiable expert credentials in a decentralized evaluation marketplace.
- Integrated three LLM providers (OpenAI, Anthropic, Gemini) with dynamic routing, picking the best model per task to cut inference spend.
- Established observability stack with OpenTelemetry, distributed tracing, and structured logging across services.
- Hired and led a distributed team of up to 3 engineers; owned technical interviews and architectural reviews.

Fractional CTO (Part-time, hands-on)

Rocking Product — Apr 2025 – Dec 2025 (part-time, parallel to KliverAI) — Remote

- Came in as Fractional CTO to an 8-engineer team and restructured it in the first quarter without stepping away from the code; ran technical interviews and hiring decisions.
- Built an image embeddings service (Azure AI Vision and CLIP) to improve product matching across brand comparisons, raising accuracy and lifting the client's revenue by 40%.
- Cut production bugs by 70% through team training and automated processes (testing, CI / CD, code review).
- Optimized cloud infrastructure and runtime systems through caching, compression, and dynamic resource selection, cutting combined spend by 45%.
- Owned architectural design end-to-end — service boundaries, data model design, technology selection — formalized through ADRs; introduced Spec-Driven Development (SDD); proposed and led a full-stack migration from a fragmented JavaScript codebase to .NET.

Lead Rust Engineer (Distributed Systems)

Borderless Capital / CTF Capital — Sep 2021 – Mar 2025 — Remote

- Shipped and operated production Rust services (tokio, async / await, Pin) processing 10K+ opportunities per day from $50M+ in daily transactional flow and contributing $100K+ in attributable annual P&L with execution under 10ms p99; tuned tokio runtime parameters, async boundaries, and lock-free coordination patterns for sub-10ms targets.
- Migrated the system from Python / Cython to Rust and took it to 20x more transactions per second.
- Built custom transaction delivery infrastructure in Rust with parallel broadcasting across multiple RPC endpoints and direct validator / TPU routes, achieving sub-slot inclusion latencies (~400ms on Solana).
- Designed async coordination across multi-chain ecosystems (EVM — Ethereum, Arbitrum, Optimism — and Solana) with custom protocol implementations, mempool analysis, and gas estimation logic.
- Shipped Python / Cython FFI bindings exposing Rust performance-critical code to higher-level orchestration layers, eliminating GC pauses across language boundaries via zero-cost abstractions.
- Reverse-engineered Geth and Reth to improve transaction simulation and gas estimation, contributing to the REVM community.
- Ran technical due diligence on 5+ projects that ended up getting funded — architecture, risk, and feasibility — acting as the CTO's technical reference on the low-level side.
- Engineered low-latency observability with Prometheus and custom Rust telemetry tracking 50+ real-time metrics, with structured tracing via the tracing crate across slot timing, validator rotation, and execution outcomes.
- Owned 24/7 production reliability and incident response across trading-critical pipelines; acted as technical lead across the blockchain team, which grew from 4 to 10 engineers — code review, mentoring, technical interviewing, and architectural decisions.

Senior Software Architect

Paramo Technology — Oct 2020 – Sep 2021 — Remote

- Owned the end-to-end architecture of a high-traffic online casino (iGaming) platform handling real-time money operations and regulatory reporting, working directly with the CTO.
- Spotted the issues slowing teams down and designed action plans that raised productivity by 40%, measured by features shipped and fewer errors.
- Designed event-driven microservices in .NET, implemented distributed tracing (OpenTelemetry, Application Insights, Datadog), and brought in Kubernetes to scale.
- Led part of the hiring and designed the technical entry exam; delivered internal training on architecture patterns, design patterns, and operational best practices.

Software Architect

CloudX (Software Consultancy) — Sep 2017 – Oct 2020 — Remote / Buenos Aires (first employee)

- Was the company's first employee; led architecture decisions across several enterprise product lines, designing microservices backends using Domain-Driven Design.
- Built a distributed system over geospatial data with PostGIS for a UK client; delivered technical training and conference talks on software architecture patterns.

Software Architect (Angular / .NET)

Majestic Hotel Group — Jan 2014 – Aug 2017 — Argentina

- Designed and shipped full-stack features for internal hotel management platforms, first as Team Lead and later as architect; led migration from AngularJS to Angular 2 and redesigned the data model toward an event-oriented architecture.

.NET Web Developer

Plus54 — 2011 – 2014 — Buenos Aires (earlier: Acsys 2009–2011, IBM 2008–2009)

- Early career: built .NET web applications, internal tooling, and mobile components end-to-end.

KEY ACHIEVEMENTS

Production Distributed Systems at Borderless Capital — Shipped production tokio-based async Rust pipelines processing $50M+ in daily transactional flow with execution under 10ms across multi-chain ecosystems.

Polkadot Builder Program Graduate — Completed Polkadot Blockchain Academy in Lisbon, Portugal (2026), covering Substrate, FRAME, XCM, and runtime internals.

80+ International Technical Talks — Conferences and meetups across LATAM and Europe on Rust, distributed systems, blockchain, and software architecture.

Net-Baires Founder — Founded and grew Argentina's largest .NET developer community to 100K+ members across 10+ years.

Microsoft MVP, Developer Technologies — 8 consecutive years (2017–present), recognized for technical content and community contributions.
Microsoft Student Partner — 5 years (2016–2021).

EDUCATION & CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert (AZ-305) — Microsoft
Microsoft Certified: Azure Associate (AZ-104 / AZ-204 level) — Microsoft
Polkadot Builder Program — Polkadot Blockchain Academy, Lisbon, Portugal (2026)
Systems Engineer — Universidad Abierta Interamericana (UAI), Buenos Aires, Argentina (2017–2021)
Microsoft MVP, Developer Technologies — Feb 2017 – Present (8 consecutive years)

OPEN SOURCE & COMMUNITY LEADERSHIP

Net-Baires (.NET Community) — Founder (2015–present). Argentina's largest .NET developer community, 100K+ members, 10+ years of monthly meetups and conferences.
Crypto Para Mortales — Founder (2023–present). Spanish-language blockchain and crypto education conference.
Patrones Para Mortales — Founder (2020–present). Educational content brand on software design patterns and architecture.
YouTube channel (youtube.com/@germankuber) — technical deep-dives on Rust, blockchain, AI, and software architecture.

SPOKEN LANGUAGES

Spanish: Native — English: Full Professional Proficiency — daily use in international remote teams